# Unitary t-design with an optimal number of parameters on a linear-optical platform

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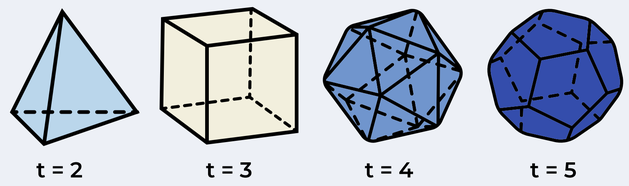
## Unitary t-design

If we have $p_t(x): \mathcal{S}(\mathbb{R}^d) \to \mathbb{R}$, with all terms homogeneous in degree at most **t**, we can define a set $X = \{x : x \in \mathcal{S}(\mathbb{R}^d)\}$ called a **spherical t-design** if $\frac{1}{|X|}\sum_{x\in X} p_t(x) = \int_{\mathcal{S}(\mathbb{R}^d)} p_t(u)d\mu_S(u)$

spherical measure




$\mathcal{S}(\mathbb{R}^3)$ examples

C.J. Colbourn. Handbook of Combinatorial Designs, 2006

If we have $p_{t,t}(U)$ with homogeneous degree at most **t** in d in the entries of a unitary matrix U and in conjugates of them, we can define a set $\mathcal{K} \subset U(d)$ called a **unitary t-design** if

$$\frac{1}{|\mathcal{K}|}\sum_{V\in\mathcal{K}} p_{t,t}(V) = \int_{U(d)} p_{t,t}(U)d\mu_H(U) \Leftrightarrow$$

replicating the first **t** moments of the Haar distribution

## Weingarten calculus

How can we numerically check a t-design?

$F^{(t)}_{\mu_H} = \int_{U(d)} |\mathrm{Tr}(U)|^{2t} d\mu_H(U)$ - frame potential (Haar) $F^{(t)}_{\mu_H} = t!,\ d \geq t$

$F^{(t)}_{\nu} = \frac{1}{|\mathcal{K}|^2}\sum_{U,V\in\mathcal{K}} |\mathrm{Tr}(UV^\dagger)|^{2t}$ - frame potential for our t-design

$F^{(t)}_{\nu} = F^{(t)}_{\mu_H}$ - exact t-design condition

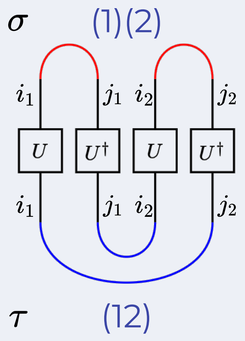
A.A. Mele. Introduction to haar measure tools in quantum information, 2024

Frame potential can be calculated via Weingarten formula

$$\int_{U(d)} U_{i_1,i_1}\dots U_{i_t,i_t}U^\dagger_{j_1,j_1}\dots U^\dagger_{j_t,j_t} d\mu_H(U) = \sum_{\sigma,\tau\in\mathcal{S}(t)} \delta_\sigma(\boldsymbol{i},\boldsymbol{j})\delta_\tau(\boldsymbol{i},\boldsymbol{j})\mathrm{Wg}(\sigma\tau^{-1}, d)$$

For $t = 2$

$\sigma$ (1)(2)




$\tau$ (12)

indexes permutation

$\delta_{i_1,j_1}\delta_{i_2,j_2}$

$\sigma\tau^{-1} = (1)(2)\cdot(12) = (12)$

$\delta_{i_1,j_2}\delta_{i_2,j_1}$

f.e. $\mathrm{Wg}((1)(2), d) = \frac{1}{d^2-1}$

Weingarten function

Inverted Gram matrix of permutations for $d \geq t$

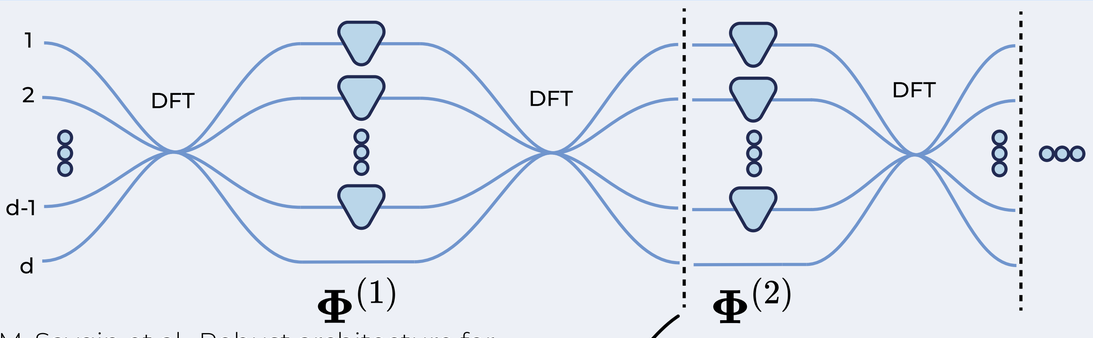
$G(\sigma\tau^{-1}, d) = d^{\#(\sigma\tau^{-1})}$

$\dim G = t! \times t!$

$\mathrm{Wg}((12), d) = -\frac{1}{d(d^2-1)}$

Chi-Fang Chen et al. Efficient unitary designs from random sums and permutations, 2024

## Optimal interferometer construction




M. Saygin et al., Robust architecture for programmable universal unitaries, 2019

Is $\mathrm{FML(t)} < \varepsilon$ ?

add one more layer!

Yes

No

$$\mathrm{FML(t)} = \left(F^{(t)}_{\nu} - F^{(t)}_{\mu_H}\right)^2$$

1. Build $|\mathcal{K}|$ matrices
2. Minimize $\mathrm{FML}(t)$

Minimum set sizes

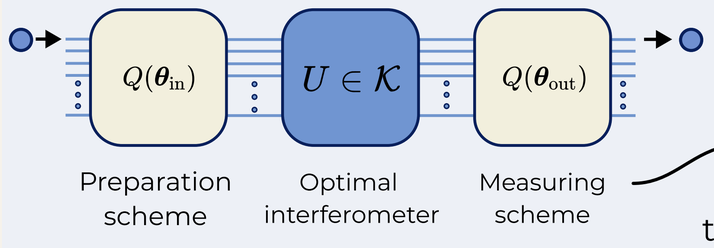
| $t$ | $\lvert\mathcal{K}\rvert$ |
|---|---|
| 1 | $d^2$ |
| 2 | $d^4 - 2d^2 + 2$ |

We found the optimal number of layers for constructing a t-design set!

Aidan Roy, A. J. Scott. Unitary designs and codes, 2008

## Optimal 1-design interferometer

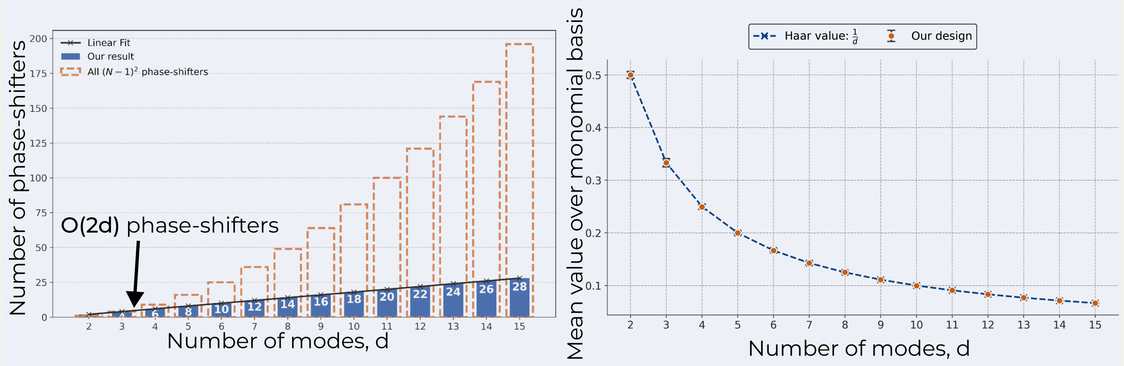
Can be modelled as a 1-photon sampling experiment




1 photon $\Leftrightarrow U(d)$

$\mathbb{E}_{\mu_H}[\circ \to \circ] = \frac{1}{d}$

Constructing a monomial basis from the matrix elements of U




Only 2 layers of Robust are needed!

## Optimal 2-design interferometer

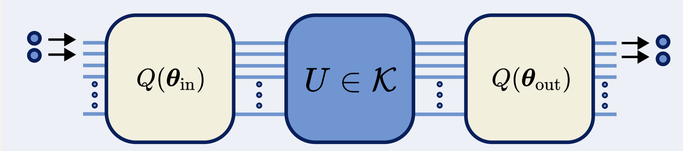
For n > 1 photons we have $\mathrm{Sym}^2\,\mathcal{H}(d) \neq U(d)$ → minimum 2 photons need to be measured!

Two photon evolution in the primary quantization picture

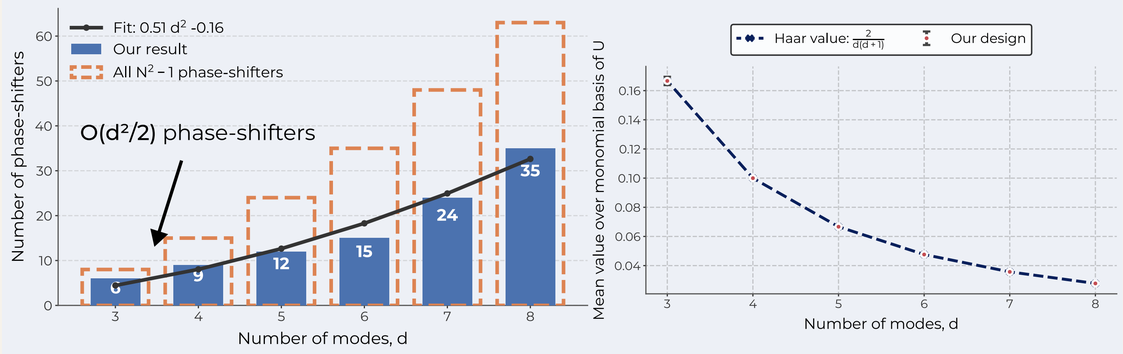
$$\mathcal{U}_2 = (U\otimes U)\,\hat{P}_{\text{sym}} \xrightarrow{\text{new potential}} \mathcal{F}^{(t)}_{\mu_H} = \frac{1}{4^t}\int_{U(d)} \left|(\mathrm{Tr}U)^2 + \mathrm{Tr}U^2\right|^{2t} d\mu_H(U)$$

So, t-design for 2-photons is equivalent to 2t-design for $U(d)$

We consider t = 1, where $\mathcal{F}^{(1)}_{\mu_H} = 1$ - from Weingarten formula



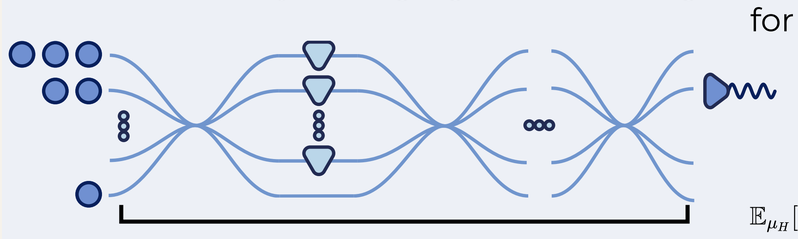

$\mathbb{E}_{\mu_H}[\circ\circ \to \circ\circ] = \frac{2}{d(d+1)}$




x2 advantage in the number of parameters!

## Sampling benchmark for 2-design

We can estimate the photon number of a Fock state with only one PNR detector by averaging over a 2-design!



k state copies optimal interferometer for 2-design

for any interferometer mode **k**

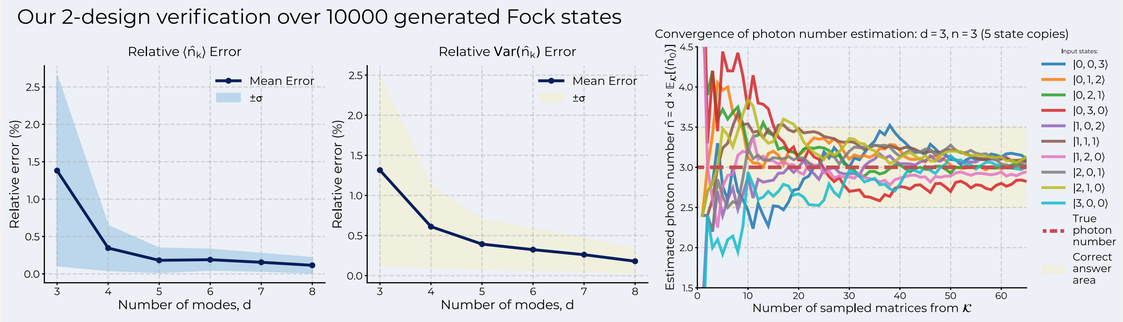
$\mathbb{E}_{\mu_H}[\langle\hat{n}_k\rangle] = \frac{n}{d}$

$\mathbb{E}_{\mu_H}[\mathrm{Var}(\hat{n}_k)] = \frac{n(d-1)(d+n)}{d^2(d+1)}$

$n = \sum_i n_i \longrightarrow n = ?$

$\mathbb{E}_{\mathcal{K}}[\cdot]$ - averaging over our 2-design

Our 2-design verification over 10000 generated Fock states

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